

-- Task 1: Identify Top Students by Total Scores

-- • Use a subquery to calculate the total score (math_score + science_score + english_score) for each student.

-- • Use an ORDER BY clause to rank students by their total scores in descending order.

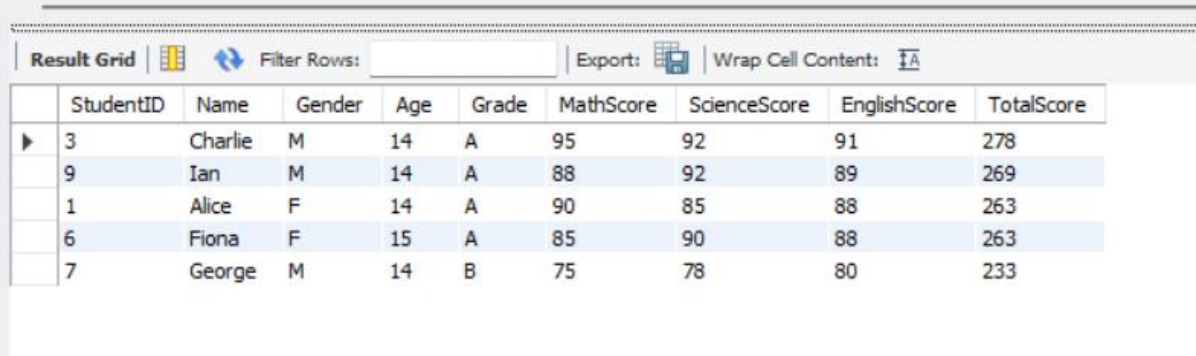
-- • Limit the results to show only the top students (e.g., top 5).

```
SELECT StudentID, Name, Gender, Age, Grade, MathScore, ScienceScore, EnglishScore,  
       (MathScore + ScienceScore + EnglishScore) AS TotalScore
```

```
FROM Students
```

```
ORDER BY TotalScore DESC
```

```
LIMIT 5;
```



	StudentID	Name	Gender	Age	Grade	MathScore	ScienceScore	EnglishScore	TotalScore
▶	3	Charlie	M	14	A	95	92	91	278
	9	Ian	M	14	A	88	92	89	269
	1	Alice	F	14	A	90	85	88	263
	6	Fiona	F	15	A	85	90	88	263
	7	George	M	14	B	75	78	80	233

❓ **SELECT Clause:**

- StudentID, Name, Gender, Age, Grade, MathScore, ScienceScore, EnglishScore: These columns are selected from the Students table.
- (MathScore + ScienceScore + EnglishScore) AS TotalScore: This calculates the total score of each student by summing their scores in Math, Science, and English, and labels the result as TotalScore.

❓ **FROM Students Clause:**

- Specifies that the data is being retrieved from the Students table.

❓ **ORDER BY TotalScore DESC Clause:**

- Orders the result in descending order based on the TotalScore, meaning the student with the highest total score appears first.

❓ **LIMIT 5 Clause:**

- Restricts the result set to only the top 5 students with the highest TotalScore.

- Calculate Averages Based on Specific Conditions
- Use subqueries to filter and group data for average calculations:
 - o Example 1: Calculate the average score of students who scored above 70 in Math.
 - o Example 2: Calculate the average total score of students grouped by a specific condition, such as a score range (e.g., students scoring 200–250 in total).

SELECT

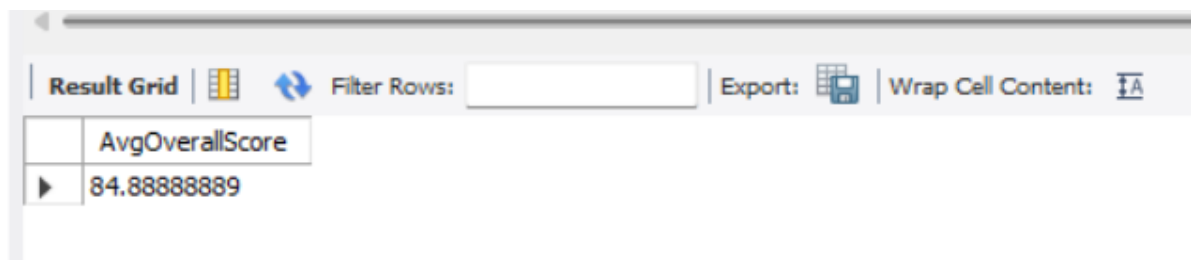
AVG((MathScore + ScienceScore + EnglishScore) / 3) AS AvgOverallScore

FROM

Students

WHERE

MathScore > 70;



The screenshot shows a database interface with a toolbar at the top containing options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below the toolbar is a table with one column labeled 'AvgOverallScore' and one row containing the value '84.88888889'.

AvgOverallScore
84.88888889

? SELECT Clause:

- AVG((MathScore + ScienceScore + EnglishScore) / 3) AS AvgOverallScore:
 - o Computes the overall score for each student by summing their MathScore, ScienceScore, and EnglishScore, then dividing by 3 to get the average subject score for each student.
 - o The AVG() function then calculates the average of these overall scores across all students who meet the condition in the WHERE clause.
 - o The result is given an alias AvgOverallScore.

? FROM Students Clause:

- Specifies that data is being retrieved from the Students table.

? WHERE MathScore > 70 Clause:

- Filters the records to include only students whose MathScore is greater than 70.
- This means only students who performed well in Math are considered in the average calculation.

SELECT

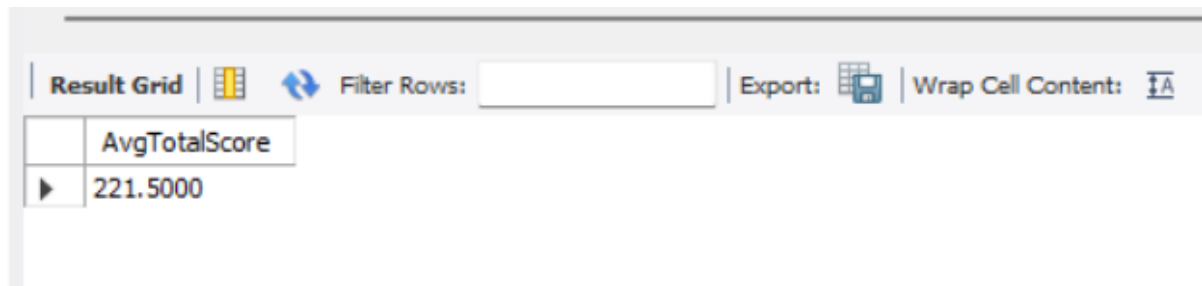
AVG(MathScore + ScienceScore + EnglishScore) AS AvgTotalScore

FROM

Students

WHERE

(MathScore + ScienceScore + EnglishScore) BETWEEN 200 AND 250;



The screenshot shows a database query result grid. The grid has two columns: 'AvgTotalScore' and a value '221.5000'. The grid is displayed in a window with a toolbar at the top containing icons for 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'.

	AvgTotalScore
▶	221.5000

❓ **MathScore + ScienceScore + EnglishScore** → Adds up the scores of three subjects for each student.

❓ **WHERE (MathScore + ScienceScore + EnglishScore) BETWEEN 200 AND 250** → Filters only those students whose total score is **between 200 and 250**.

❓ **AVG(MathScore + ScienceScore + EnglishScore) AS AvgTotalScore** → Computes the **average** of these filtered students' total scores.

-- Find Second-Highest Math Scores

-- ● Use a subquery to determine the highest Math score and exclude it in a second query to

-- find the next highest value.

-- ● Example:

-- ○ Use MAX(math_score) in a subquery to find the highest score.

-- ○ Use WHERE math_score < (SELECT MAX(math_score) FROM

-- Students) to exclude the top score and then use MAX again to find the second

-- highest score.

```

SELECT
    Name, MathScore
FROM
    Students
WHERE
    MathScore = (SELECT
        MAX(MathScore)
    FROM
        Students
    WHERE
        MathScore < (SELECT
            MAX(MathScore)
        FROM
            Students));

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Name	MathScore			
▶	Alice	90			

```

SELECT
    Name, MathScore
FROM
    Students
WHERE
    MathScore = (subquery);

```

- This retrieves the Name and MathScore of the student(s) whose MathScore matches the result from the subquery.

2. First Subquery (Finding the Highest Math Score)

sql

CopyEdit

SELECT

MAX(MathScore)

FROM

Students;

- This retrieves the maximum MathScore in the Students table.
-

3. Second Subquery (Finding the Second-Highest Math Score)

sql

SELECT

MAX(MathScore)

FROM

Students

WHERE

MathScore < (SELECT MAX(MathScore) FROM Students);

- This retrieves the maximum MathScore that is **less than** the highest MathScore, effectively finding the **second-highest** MathScore.